**1. Question**

**Write a Java program to demonstrate exception handling using ArrayStoreException when an incorrect type of value is stored in an array.**

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**2. Steps for Execution**

1. Open a Java IDE or command prompt.
2. Create a package named javapractice.
3. Create a class named StoreException.
4. Create an Object array using new String[2].
5. Use a try block to store an integer value in the array.
6. Since the actual array can store only String values, an ArrayStoreException occurs.
7. Handle the exception using the catch block.
8. Display the appropriate error message.
9. Compile and run the program.

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**3. Algorithm**

1. **Start**
2. Create an Object reference to a String array of size 2.
3. Begin the try block.
4. Display "Storing value in array".
5. Try to store the integer value 100 in arr[0].
6. Since the actual array is a String array, an ArrayStoreException occurs.
7. Catch the exception using catch.
8. Display "Wrong type of value stored in array".
9. **Stop**

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**4. UML Class Diagram**

```
┌──────────────────────────────────┐
|          StoreException          |
├──────────────────────────────────┤
|                                  |
├──────────────────────────────────┤
| + main(args : String[]) : void   |
└──────────────────────────────────┘
              |
              | catches
              ▼
┌──────────────────────────────────┐
|        ArrayStoreException       |
├──────────────────────────────────┤
|                                  |
| Exception when an incompatible value |
| is stored in an array            |
└──────────────────────────────────┘
```

**5. Code**

```
package javapractice;

public class StoreException {
    public static void main(String[] args) {
        Object arr[] = new String[2];

        try {
            System.out.println("Storing value in array");
            arr[0] = 100;
        }
```

```
        catch (ArrayStoreException e) {
            System.out.println("Wrong type of value stored in array");
        }
    }
}
```

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**7. Output**

Storing value in array

Wrong type of value stored in array

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**8. Result**

**Thus, the Java program successfully demonstrates the handling of ArrayStoreException when an incompatible data type is stored in an array.**